

Lab 1 Assignment

Peter James

2025-04-17

Load Packages

```
library(sf)
library(tidyverse)
library(tidygeocoder)
library(MapGAM)
library(tidycensus)
library(flextable)
library(tmap)
```

Geocoding

Question 1. Read in In N Out Burger

```
glimpse(innout)
```

```
download.file("https://raw.githubusercontent.com/pjames-ucdavis/SPH215/refs/heads/main/innout_final.csv")
in_n_out <- read_csv("innout_final.csv")
```

```
## Rows: 30 Columns: 2
## -- Column specification -----
## Delimiter: ","
## chr (1): store address
## dbl (1): Store Number
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Question 2. Glimpse at the data. Looks like *store address* is the variable to geocode!

```
glimpse(in_n_out)
```

```
## Rows: 30
## Columns: 2
## $ `Store Number` <dbl> 131, 327, 276, 217, 225, 247, 320, 319, 197, 207, 143, 180, 308, 141, 89, 13
## $ `store address` <chr> "3501 Truxel Road Sacramento CA 95834", "8200 Delta Shores Circle Sacramento
```

Question 3. Geocode the data using “arcgis” geocoder.

```
innout_geo      <- geocode(in_n_out, address = "store address",
                           method = "arcgis")
```

```
## Passing 30 addresses to the ArcGIS single address geocoder
```

```
## Query completed in: 13.3 seconds
```

Question 4. Looks like there's no missingness! No NAs!

```
summary(innout_geo$lat)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   35.99  37.99   38.39   38.12  38.59   38.76
```

```
summary(innout_geo$long)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  -122.7  -122.0  -121.5  -121.5  -121.3  -119.7
```

Question 5. We use `st_as_sf` to make the dataset spatial and we use the crs of 4326

```
innout_pts <- st_as_sf(innout_geo, coords=c("long", "lat"), crs=4326)
```

Question 6. Map the addresses in red below!

```
tmap_mode("plot")
```

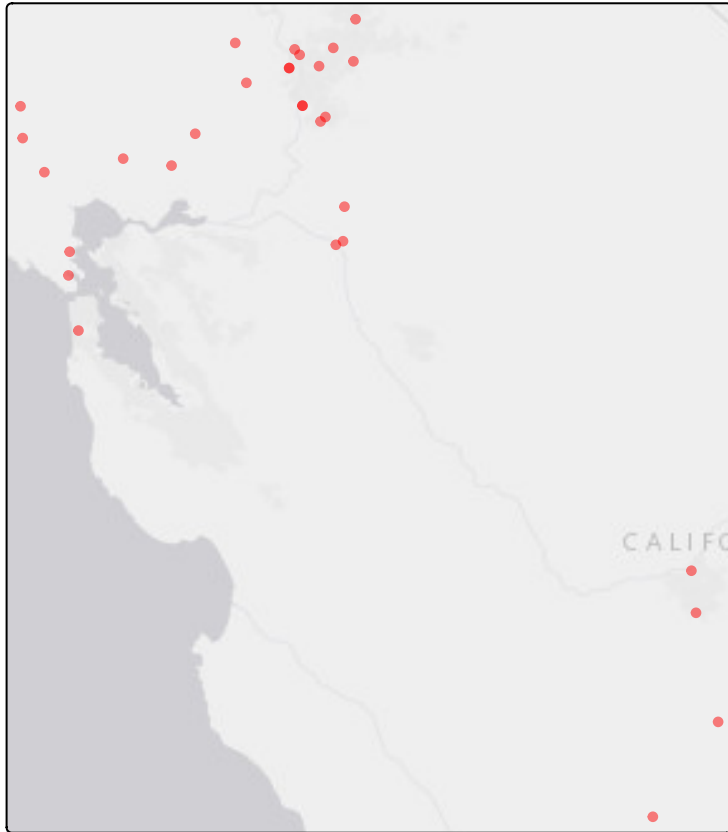
```
## i tmap modes "plot" - "view"
```

```
innout_map = tm_shape(innout_pts) + tm_dots(col = "red", size = 0.3, alpha = 0.5) + tm_basemap()
```

```
##
```

```
## -- tmap v3 code detected -----
## [v3->v4] `tm_dots()`: use `fill_alpha` instead of `alpha`.
```

```
innout_map
```

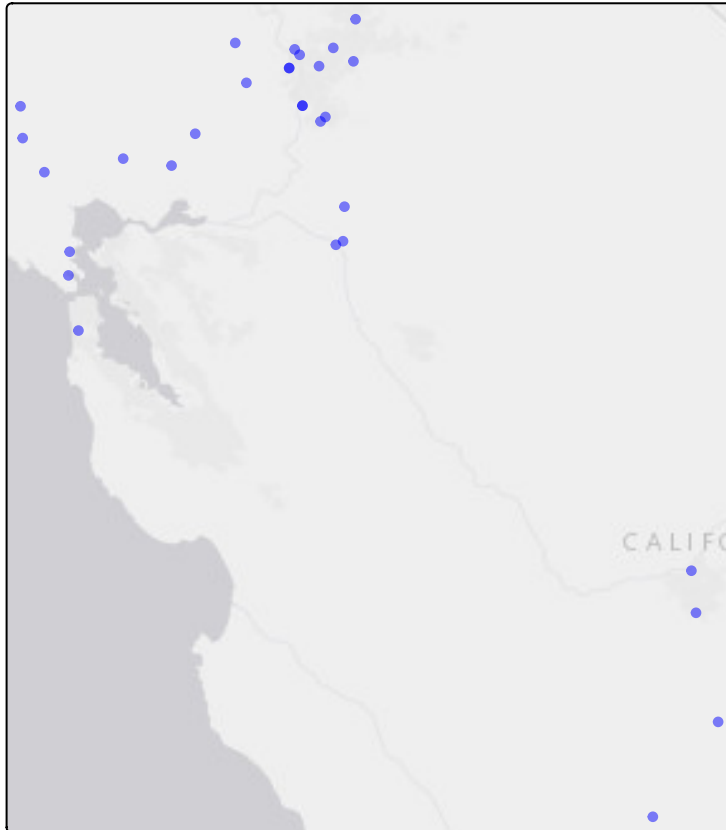


Question 7. Map the addresses in blue below!

```
tmap_mode("plot")

## i tmap modes "plot" - "view"
innout_map_blue = tm_shape(innout_pts) + tm_dots(col = "blue", size = 0.3, alpha = 0.5) + tm_basemap()

##
## -- tmap v3 code detected -----
## [v3->v4] `tm_dots()`: use `fill_alpha` instead of `alpha`.
innout_map_blue
```



Map Point Health Data

Question 8. Import CAdat dataset

#Load CAdat dataset from MapGAM package

```
data(CAdat)
ca_pts <- CAdat
summary(ca_pts)
```

##	time	event	X	Y	AGE	INS
##	Min. : 0.004068	Min. : 0.0000	Min. : 1811375	Min. : -241999	Min. : 25.00	Mcd: 431
##	1st Qu.: 1.931247	1st Qu.: 0.0000	1st Qu.: 2018363	1st Qu.: -94700	1st Qu.: 53.00	Mcr: 1419
##	Median : 4.749980	Median : 1.0000	Median : 2325084	Median : -60386	Median : 62.00	Mng: 2304
##	Mean : 6.496130	Mean : 0.6062	Mean : 2230219	Mean : 87591	Mean : 61.28	Oth: 526
##	3rd Qu.: 9.609031	3rd Qu.: 1.0000	3rd Qu.: 2380230	3rd Qu.: 318280	3rd Qu.: 71.00	Uni: 168
##	Max. : 24.997764	Max. : 1.0000	Max. : 2705633	Max. : 770658	Max. : 80.00	Unk: 152

```
ca_pts <- st_as_sf(CAdat, coords=c("X", "Y"))
```

Question 9. Add CRS

#Load the projection into an object called ca_proj

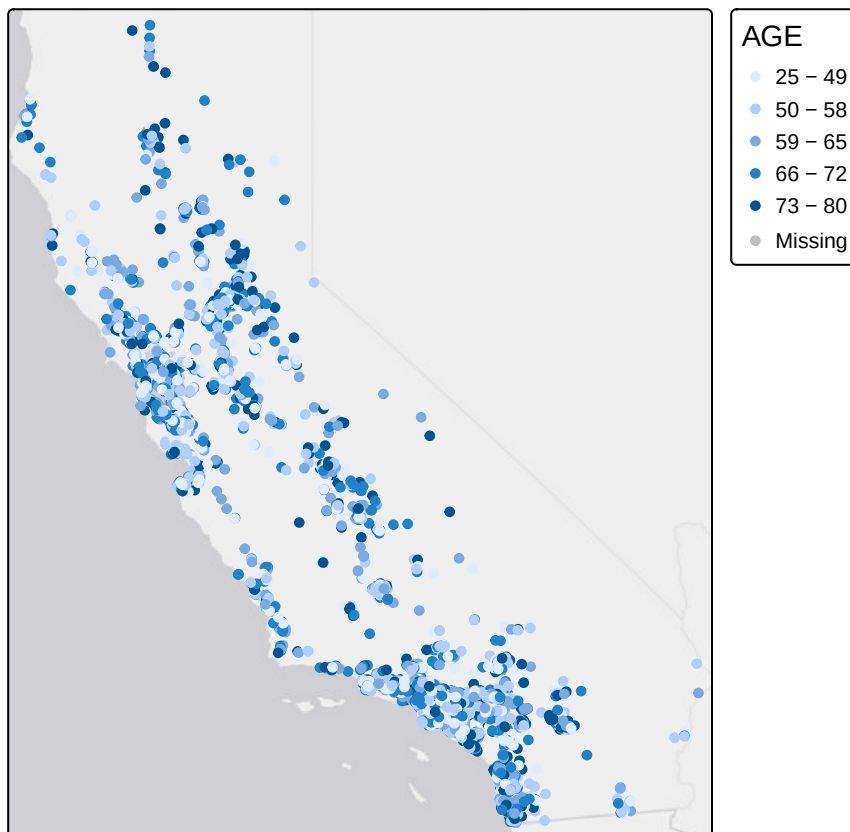
```
ca_proj <- "+proj=lcc +lat_1=40 +lat_2=41.666666666666666
+lat_0=39.333333333333334 +lon_0=-122 +x_0=2000000
+y_0=500000.0000000002 +ellps=GRS80
+datum=NAD83 +units=m +no_defs"
```

```
#Set CRS
ca_pts_crs <- st_set_crs(ca_pts, ca_proj)
```

Question 10. Create a map of the CAdata dataset with points color coded by quintiles of age.

```
cancer_map_age = tm_shape(ca_pts_crs) + tm_dots(col = "AGE", size = 0.3, style = "quantile") + tm_basemap()

##
## -- tmap v3 code detected -----
## [v3->v4] `tm_dots()`: instead of `style = "quantile"`, use fill.scale = `tm_scale_intervals()`.
## i Migrate the argument(s) 'style' to 'tm_scale_intervals(<HERE>)'
cancer_map_age
```

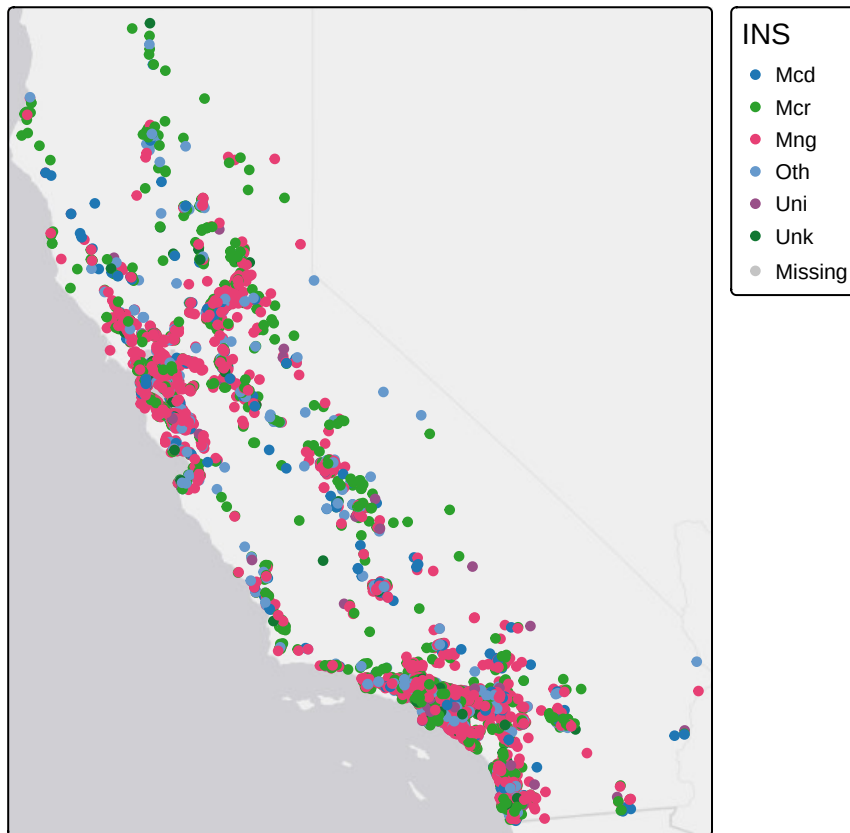


Question 11. Map by insurance status

```
cancer_map_ins = tm_shape(ca_pts_crs) + tm_dots(col = "INS", size = 0.3, style = "cat") + tm_basemap()

##
## -- tmap v3 code detected -----
## [v3->v4] `tm_dots()`: instead of `style = "cat"`, use fill.scale = `tm_scale_categorical()`.
cancer_map_ins
```

```
cancer_map_ins
```



Census Data

Question 12. Use the tidycensus package to download American Community Survey Five year estimates from 2018-2022 in California at the Census tract level for the following variable

```
ca <- get_acs(geography = "tract",  
              year = 2022,  
              variables = c(medinc = "B19013_001"),  
              state = "CA",  
              survey = "acs5",  
              output = "wide")
```

```
## Getting data from the 2018-2022 5-year ACS
```

```
## Using FIPS code '06' for state 'CA'
```

Question 13. Take a glimpse at your data.

```
glimpse(ca)
```

```
## Rows: 9,129
```

```
## Columns: 4  
## $ GEOID    <chr> "06001400100", "06001400200", "06001400300", "06001400400", "06001400500", "06001400600"  
## $ NAME      <chr> "Census Tract 4001; Alameda County; California", "Census Tract 4002; Alameda County;  
## $ medincE   <dbl> 234236, 225500, 164000, 158836, 95078, 155893, 110556, 97132, 114583, 153654, 93121,  
## $ medincM   <dbl> 42845, 29169, 44675, 9038, 34125, 33824, 12318, 37490, 31849, 31601, 8351, 41439, 12
```

Question 14. Give us the mean median household income across tracts in California.

```
ca %>%  
  summarize(Mean = mean(medincE))
```

```
## # A tibble: 1 x 1
##   Mean
##   <dbl>
## 1    NA
```

Question 15. Make a table showing the mean, median, and standard deviation of median household income across tract in California, and make it look fancy!

```
summarytable <- ca %>%
  summarize(Mean = mean(medincE),
            Median = median(medincE),
            SD = sd(medincE)
            )
my_table <- flextable(summarytable)
my_table
```

```
## Warning: fonts used in `flectable` are ignored because the `pdflatex` engine is used and not `xelatex`  
## can avoid this warning by using `set_flectable_defaults(fonts_ignore=TRUE)` or use a compatible engine  
## xelatex`).
```

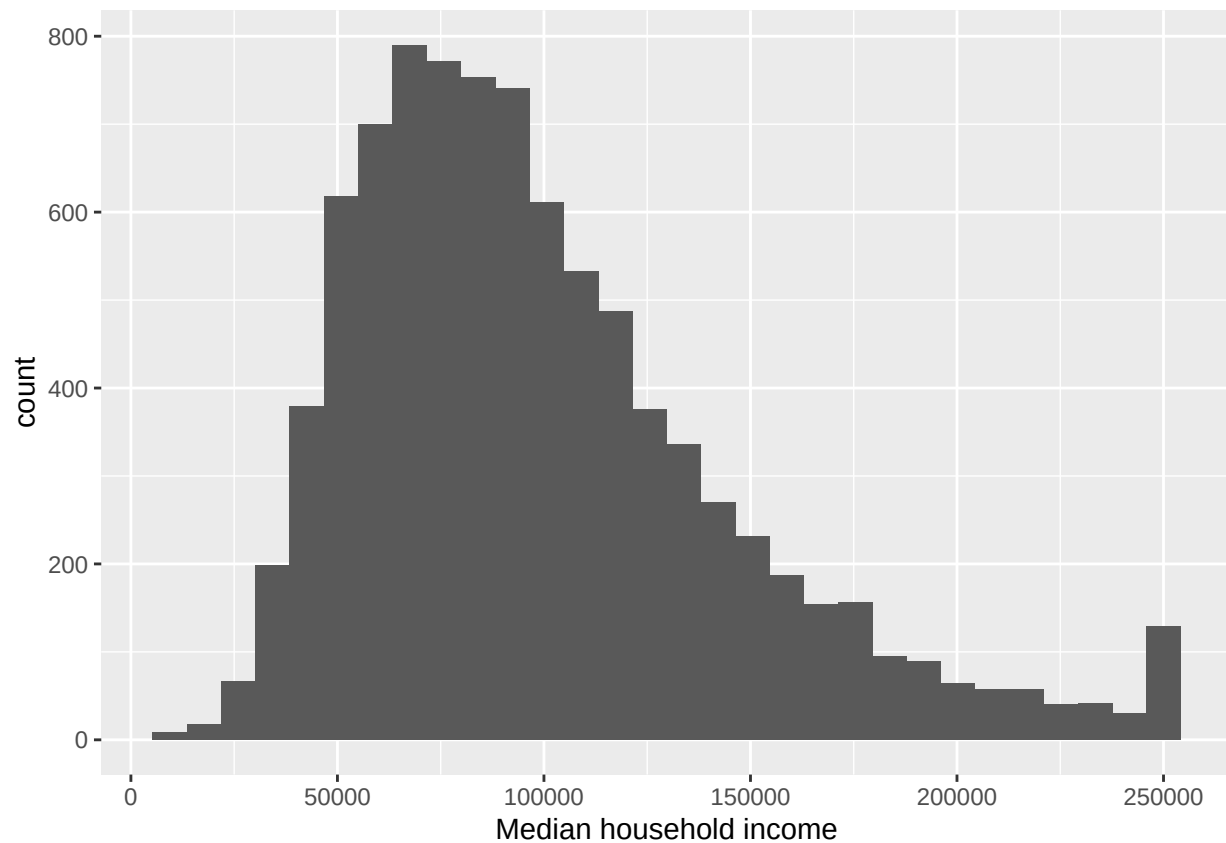
Mean	Median	SD
------	--------	----

Question 16. Create a histogram of median household income across tracts in California. Make sure it has a label.

```
ca %>%
  ggplot() +
  geom_histogram(mapping = aes(x=medincE)) +
  xlab("Median household income")
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

```
## Warning: Removed 141 rows containing non-finite outside the scale range (`stat_bin()`).
```



Question 17. OK now give me a boxplot! And I want a nice label for this too!

```
ca %>%  
  ggplot() +  
    geom_boxplot(mapping = aes(y = medincE)) +  
    ylab("Median household income")
```

```
## Warning: Removed 141 rows containing non-finite outside the scale range (`stat_boxplot()`).
```

